Acyclovir kinetics in end-stage renal disease

Acyclovir (ACV) is almost entirely eliminated by the kidneys and has a terminal plasma half-life (t½) of 2 to 3 hr in subjects with normal renal function. To determine the drug's kinetics and tolerance in patients with severe renal failure, six anuric subjects on long-term hemodialysis were studied. Each received a 1-hr infusion of 2.5 mg/kg IV ACV. The kinetics are well described by a two-compartment open model. ACV terminal plasma t½ and the total body clearance were 19.5 ± 5.9 hr ($\bar{x}$ ± SD) and 28.6 ± 9.5 ml/min/1.73 m². Peak (end of infusion) and 8- and 24-hr plasma ACV concentrations were 37.5 ± 23.3, 10.3 ± 2.9, and 6.4 ± 2.4 μM. Approximately 48 hr after the start of the infusion the subjects were hemodialyzed for 6 hr. The pre- and posthemodialysis ACV plasma levels were 2.74 ± 1.38 and 1.11 ± 0.60 μM. The terminal ACV t½ during hemodialysis was 5.7 ± 0.85 hr. During hemodialysis paired arterial and venous samples showed that ACV was readily dialyzed, with a mean coefficient of extraction of 0.45 ± 0.12. The dialysis clearance of acyclovir was 81.8 ± 12.6 ml/min. None of the patients had any ACV-related adverse effects. Since ACV elimination is markly reduced in end-stage renal failure and because ACV is readily hemodialyzible, dosage modifications are needed to avoid cumulation and to replace dialyzed drug.

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Acyclovir [9-(2-hydroxyethoxymethyl)guanine, Zovirax] (ACV) is currently being investigated for use in herpesvirus infections,[3, 11, 17, 18] particularly those caused by herpes simplex and varicella zoster virus. ACV is an acyclic nucleoside analog of guanosine. It has selective activity in cells infected with herpesvirus by virtue of specific phosphorylation by the viral-coded deoxynucleoside kinase and the selectivity of ACV triphosphate both as a substrate and as an inhibitor of the herpes DNA polymerase.[7, 9] This selectivity accounts for its lack of toxicity to mammalian cells, laboratory animals, and humans at levels that far exceed the inhibitory dose (ID_{50}) for susceptible herpesviruses in vitro.[4–6, 17, 18]

ACV has been shown to be efficacious in

Supported by a grant from the Burroughs Wellcome Co. Patients were studied at the Johns Hopkins General Clinical Research Center, which is supported by the Division of Research Resources (5M01RR00035), National Institutes of Health.

Presented, in part, at the International Acyclovir Symposium, National Institutes of Health, Bethesda, Md.

Received for publication Dec. 8, 1981.

Accepted for publication Dec. 19, 1981.

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Table I. *Demographic data*

Patient No.	*Sex*	*Age (yr)*	*Weight (kg)*	*Height (cm)*	*BSA* (m^2)*
1	F	72	74.0	173	1.90
2	M	30	57.0	172	1.65
3	M	45	48.0	173	1.52
4	M	30	52.9	173	1.60
5	F	63	50.1	163	1.51
6	M	35	63.2	185	1.80
$\bar{x} \pm$ SD		46 ± 18	57.5 ± 9.7	173 ± 7	1.66 ± 0.17

*Body surface area (in m^2) = $0.0235 \cdot H^{0.422} \cdot W^{0.515}$ where height (H) is in centimeters and weight (W) is in kilograms.

vitro and in vivo in preclinical studies.[2–4, 8, 14, 18] In man, topical ACV has been shown to be effective in the treatment of herpetic keratoconjunctivitis.[11] In a study done in the United Kingdom, parenteral ACV seemed to be able to arrest infections caused by herpesviruses in 23 patients; it seemed to be most effective when given early.[19] In a randomized, double-blind, placebo-controlled trial, ACV was very effective in preventing herpes simplex virus I reactivation in bone marrow transplant recipients.[17]

In single-dose studies the terminal plasma half-life ($t_{1/2}\beta$) of ACV was found to be approximately 3 hr in patients with cancer and normal renal function.[1, 5, 13, 20] The major site of ACV elimination is the kidney. The only significant metabolite that has been identified to date is 9-carboxymethoxymethylguanine.[5, 6] ACV renal clearance after a single 1-hr infusion was two to three times that of creatinine clearance.[5, 6, 13] This suggests that the drug is eliminated by renal tubular secretion in addition to glomerular filtration and also that a drug with the kinetic profile of ACV would be greatly influenced by diminished renal function. Our study was conducted to investigate the kinetics and tolerance of ACV between and during hemodialyses in subjects with end-stage renal failure.

Material and methods

Our subjects were six patients (four men and two women) who were over 18 yr old and who were followed at The Johns Hopkins Dialysis Unit. All had end-stage renal failure and 4 were surgically anephric. All were anuretic. Their demographic data are listed in Table I.

ACV was provided by Burroughs Wellcome as the sterile lyophilized sodium salt. Each 50-ml vial containing the equivalent of 500 mg ACV was reconstituted with 25 ml sterile water for injection. The calculated dose for each patient was withdrawn and enough 5% dextrose in water was added to bring the final volume to 60 ml. This was given by constant intravenous infusion over 1 hr (1 ml/min) through a peripheral vein. All the subjects were given 2.5 mg/kg on the morning after a scheduled dialysis.

Venous blood samples for ACV levels were drawn through an indwelling catheter inserted in the arm contralateral to their fistula site. Samples were placed in 5-ml heparinized tubes. ACV levels were determined on samples drawn before infusion, at 1 hr (end of the infusion), and 1.5, 2, 3, 4, 6, 8, 10, 12, 16, 24, 32, 40, and 48 hr after the start of the infusion. Patients then underwent hemodialysis for 6 hr. Simultaneous paired arterial and venous samples were obtained at 5 min and 1, 2, 3, 4, and 5 hr into hemodialysis and at the end of dialysis (6 hr). A final sample was drawn 0.25 to 1.5 hr after the end of dialysis. Blood samples were centrifuged and the plasma was removed and frozen for subsequent ACV determination.

All histories were taken and physical and laboratory evaluations performed on admission of patients to the Johns Hopkins Hospital Clinical Research Unit. After the ACV infusion potential adverse clinical effects were monitored daily for 3 days and again 3 wk later. Laboratory evaluations, including tests of hematologic, renal, and liver function were followed frequently during the 4-day hospitalization and then again at the follow-up visit.

Plasma ACV concentrations were determined

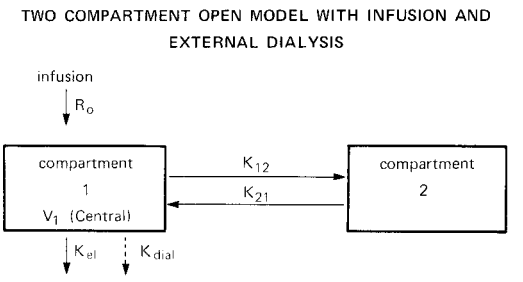


Fig. 1. Kinetic model. $k_{dial} = 0$ in the interdialysis period.

by a sensitive and specific radioimmunoassay. The assay was performed by the ammonium sulfate precipitate method.[16] The lower limit of detection of this assay is 0.05 μM.

Kinetic analysis. Visual inspection of the plasma concentration values plotted as a function of time indicated that the concentrations appeared to decay in a biexponential manner after infusion and also during the dialysis period. This pattern is consistent with a two-compartment open model. The individual ACV plasma profiles were model-fitted using the biexponential decay function previously described.[5] The model was modified to incorporate hemodialysis by including a second elimination route from the first compartment parallel to the usual one described by k_{el} (Fig. 1). Its first-order rate constant is called k_{dial}.

Parameter estimation using the data from both interdialysis and dialysis periods requires fitting a discontinuous function, the discontinuity occurring at the initiation of dialysis. In our study it was assumed that the processes governing ACV distribution during dialysis were the same as before dialysis, with the exception of the additional elimination route for the hemodialyzer from the first compartment. This means that none of the kinetic parameters except the elimination rate constant for dialysis (k_{dial}) (and the α and β values dependent on it) changed from their predialysis values during the course of hemodialysis. The k_{dial} value remained constant throughout the hemodialysis period. As is required for implementation of these assumptions, the parameter estimation routine ensured that the final conditions of the interdialysis period and the initial conditions of the dialysis period were identical, i.e., the amounts of ACV in the first and second compartments at the end of the interdialysis period equaled the amounts at the start of hemodialysis.

For each patient the values of the elimination rate constant (k_{el}), the k_{dial}, the interdialysis rate constants of the rapid and slow disposition phase (α and β), the terminal disposition rate constant of dialysis ($D\beta$), the intercompartmental rate constants from the first (central) compartment to the second (peripheral) compartment and from the second compartment to the first (k_{12} and k_{21}), and the volume of distribution of the central compartment (V_1) were obtained by a nonlinear least-squares parameter estimation routine based on the Newton-Raphson method.[15] The interdialysis half-life of the rapid disposition phase ($t\frac{1}{2}\alpha$), the $t\frac{1}{2}\beta$, and the terminal $t\frac{1}{2}$ of dialysis ($t\frac{1}{2}D\beta$) were calculated from their respective rate constants. The volume of distribution at steady state (Vdss), total body clearance (Cl_{tot}), and the dialysis clearance (Cl_{dial}) were calculated using the following equations:

$$Vd_{ss} = V_1 \cdot \frac{k_{12} + k_{21}}{k_{21}} \quad (1)$$

$$Cl_{tot} = k_{el} \cdot V_1 \quad (2)$$

$$Cl_{dial} = k_{dial} \cdot V_1 \quad (3)$$

The body surface area of each patient was estimated by the use of a relationship based on weight and height.[10] The ACV coefficient of extraction (E_c) during hemodialysis was calculated as:

$$E_c = (A\text{-}V)/A \quad (4)$$

where A is the concentration of ACV in arterial plasma and V is the ACV concentration in the plasma leaving the dialyzing coil.

Results

ACV concentrations. Subjects' plasma ACV concentration-time data plots are shown in Fig. 2 and demonstrate that the concentration-time profiles, both between and during hemodialysis, are well described by a two-compartment open kinetic model. After a 1-hr infusion, mean (±SD) ACV plasma concentrations (Fig. 3) at 1 hr (peak), 8 hr, and 24 hr were 37.5 ± 23.3, 10.3 ± 2.9, and 6.4 ± 2.4 μM and ACV was found to persist in the plasma for more than

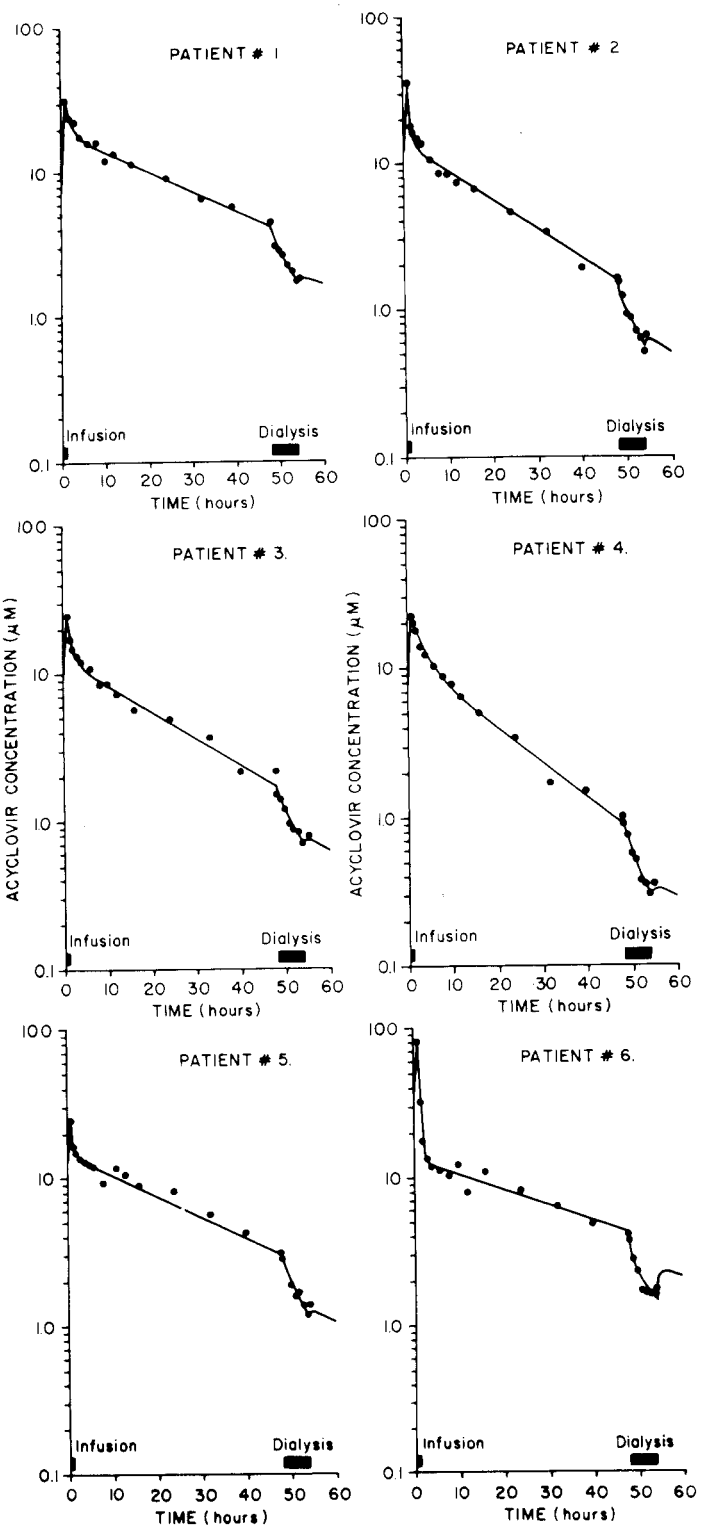


Fig. 2. Individual plasma ACV concentration-time profiles. Plasma ACV levels are (•) after a 1-hr ACV 2.5-mg/kg infusion. The line represents the best-fit biexponential equation for each subject.

Table II. *Kinetic parameters*

Patient No.	*t½α (hr)*	*t½β (hr)*	*k_{12} (l/hr)*	*k_{21} (l/hr)*	*k_{el} (l/hr)*
1	0.98	21.70	0.300	0.383	0.059
2	0.30	15.04	1.650	0.500	0.214
3	0.65	16.67	0.563	0.448	0.099
4	1.81	13.02	0.133	0.203	0.100
5	0.25	21.28	1.900	0.812	0.112
6	0.29	29.16	1.940	0.192	0.299
$\bar{x}$ ± SD	0.71 ± 0.61	19.48 ± 5.85	0.914 ± 0.801	0.423 ± 0.229	0.147 ± 0.091

Table III. *Plasma ACV concentrations, arterial-venous (A-V) differences, and E_cs at indicated times during hemodialysis*

Time (hr)*	*ACV Conc (arterial) (μM)*	*ACV Conc (venous) (μM)*	*A-V difference (μM)*	*E_c*
0.1	2.10 ± 1.16	1.20 ± 0.50	0.91 ± 0.67	0.40 ± 0.08
1.0	1.86 ± 0.90	1.08 ± 0.71	0.78 ± 0.67	0.43 ± 0.24
2.0	1.61 ± 0.86	0.93 ± 0.56	0.68 ± 0.41	0.42 ± 0.10
3.0	1.37 ± 0.76	0.77 ± 0.42	0.60 ± 0.34	0.44 ± 0.03
4.0	1.25 ± 0.71	0.63 ± 0.40	0.62 ± 0.35	0.50 ± 0.07
5.0	1.14 ± 0.63	0.53 ± 0.31	0.61 ± 0.35	0.53 ± 0.07
6.0	1.00 ± 0.58	0.5 ± 0.32	0.38 ± 0.26	0.41 ± 0.12

*From start of hemodialysis.
All values are mean ± SD.

48 hr. Mean plasma concentrations of the drug before and after dialysis were 2.74 and 1.11 μM. During hemodialysis plasma ACV concentrations declined more rapidly than in the interdialysis period. After hemodialysis the ACV plasma concentrations rose slightly in each subject, presumably due to the reestablishment of a new "equilibrium" between the peripheral and central compartments following the discontinuation of dialysis.

Kinetic analysis. The kinetic parameters computed for each patient are presented in Table II. The mean (±SD) terminal ACV plasma t½s between and during hemodialyses were 19.5 ± 5.9 and 5.7 ± 0.85 hr. The V_1 was 15.3 *l*/1.73 m^2; there was, however, a considerable degree of interpatient variability in this parameter (SD = 8.1 *l*/1.73 m^2). One patient (No. 6) had a very small central compartment (3.6 *l*/1.73 m^2) and the same patient had the highest peak concentration (83.9 μM, Fig. 2). ACV Cl_{tot} was 28.6 ± 9.5 ml/min/1.73 m^2. During hemodialysis ACV Cl_{dial} was 81.8 ± 12.6 ml/min. This clearance requires a mean blood flow of 225 ml/min through the dialyzer (based on the subject's Cl_{dial}, E_c, and hematocrit), which is on the same order as the usual dialysis blood flow of approximataly 200 ml/min used in our dialysis unit.

Table III shows mean arterial and venous differences and mean E_cs during hemodialysis. As shown, E_c remained constant during dialysis. The average E_c was 0.45 ± 0.12 (n = 40). After dialysis drug concentrations were approximately 40% of those before dialysis.

Toxicity. None of the patients had adverse effects that were evident either clinically or by laboratory evaluation and that could be attributed to ACV. Local intravenous site toxicity was not observed.

Discussion

ACV is a potent and selective antiviral drug with widespread application in the treatment-

V_1 (l/1.73m²)	Vdss (l/1.73m²)	Cl_{tot} (ml/min/1.73 m²)	k_{dial} (l/hr)	Cl_{dial} (ml/min)	$t½D_\beta$ (hr)
20.80	37.10	20.5	0.191	72.6	5.90
9.73	41.86	34.7	0.588	91.1	4.86
19.50	44.00	32.2	0.234	66.7	5.68
25.49	42.19	42.5	0.208	81.5	5.82
12.49	41.73	23.3	0.427	77.7	4.93
3.63	40.31	18.1	1.607	101.2	7.19
15.27 ± 8.08	41.20 ± 2.33	28.6 ± 9.5	0.543 ± 0.544	81.8 ± 12.6	5.73 ± 0.85

of various herpetic infections. Others have shown,[5, 6, 13, 20] in patients with normal renal function, that ACV is eliminated predominantly by the kidneys. The only significant metabolite isolated in these patients was carboxymethoxymethylguanine, which accounted for less than 14% of the dose.[5, 6] As anticipated in patients with end-stage renal disease, ACV elimination was significantly reduced. Cl_{tot} of ACV between dialyses was approximately 10% of that reported for patients without renal insufficiency.[1, 5, 6, 13] ACV plasma $t½\beta$ was about seven times that in patients with normal renal function.[1, 5, 6, 13, 20] In our patients with end-stage renal disease ACV could be detected ($\geq$1 μM) even after 48 hr. This is in contrast to subjects with normal renal function given the same dose of drug in an identical fashion; their ACV concentrations were under 1 μM at 11 hr and were not detectable ($<$0.05 μM) at 36 hr.[5, 13]

Since ACV is a low molecular weight compound (225), is relatively soluble in water (maximum solubility = 5556 μM), and is only 9% to 22% bound to plasma proteins,[6] it was expected to be hemodialyzable. The E_c was 0.45 and a single 6-hr hemodialysis lowered the mean ACV plasma concentration by approximately 60%. The Cl_{dial} was found to be 82 ml/min, which trebled the Cl_{tot} in these patients when not on hemodialysis. This Cl_{dial} estimate based on the decrease in plasma concentration in all subjects is approximately the calculated value of 113 ml/min, which was based on the appearance of ACV in the dialysate (available for patients 1, 2, and 3 only).[12]

When ACV is given to patients with end-stage renal failure, dosage modification will be necessary to maintain concentrations at an effective level without concomitant cumulation. The effective level has not yet been determined; it could be the ED_{50} or ED_{99} for the virus or it may be some other concentration that inhibits viral replication.

The steady-state ACV concentrations will depend on both the maintenance dose and the dosage interval chosen and modifications can be achieved by changing either of these. For example, a loading dose could be given that would enable a therapeutic level to be reached and smaller maintenance doses could then be given every 8 hr.

Computer simulations using the two-compartment model discussed earlier indicate that a loading dose of 37% and an every-8-hr maintenance dose of 14% of the recommended dose for a similar patient with normal renal function should induce and maintain the same mean plasma concentrations. An advantage of this schedule is that it minimizes fluctuations between peak and trough concentrations. This advantage is only theoretical, however, since neither toxic nor therapeutic concentrations have been established.

An example of changing the dosage interval to attain steady state would be to give a full standard ACV dose every second or third t½. In the anuretic patient this would be about every 48 hr. An advantage of changing the interval is that the drug is given less frequently, which may be useful where venous access is a problem. This regimen, however, results in higher peak and lower trough plasma concentrations and, if the critical ACV concentration that is necessary to inhibit viral replication is higher than trough levels, the schedule may result in long periods of inadequate concentrations. There is no evidence

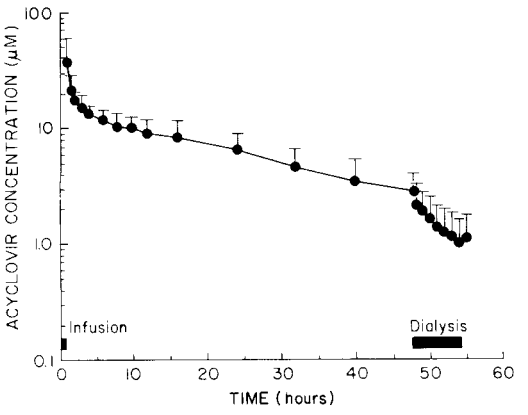


Fig. 3. Mean (±SD) plasma ACV concentration-time profile after a 1-hr infusion of 2.5 mg/kg in the interdialysis and dialysis period.

yet that one schedule would be more efficacious or less toxic than the other; either should result in approximately the same mean plasma concentrations for anuretic patients.

During hemodialysis about 60% of the drug in the body will be removed and should be replaced. Depending on the time dialysis occurs in relation to the preceding dose, it will be 60% to 100% of the full loading dose. For example, if dialysis is performed immediately after dosing, it would be reasonable to give 60% of a loading dose after dialysis. If dialysis is performed at the end of a dosing interval (when the amount of drug present in the body is already reduced), then it would be reasonable to repeat the loading dose.

We gratefully thank Dr. Gertrude B. Elion and Dr. S. H. T. Liao for their helpful suggestions and comments in the preparation of this manuscript.

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